



# Sheltered by trees: long-term yield dynamics in temperate alley cropping agroforestry with changing water availability

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Accepted: 2 April 2025 / Published online: 20 May 2025  
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## Abstract

As warm season droughts increase in frequency due to climate change, causing severe yield losses especially among cereal crops, European agriculture is in dire need of adaptation. While agroforestry is widely regarded as a key adaptation measure, little is known on how yield performance is influenced by changing water availability in temperate regions. Therefore, we assessed the yield dynamics of five winter crops (winter wheat, triticale, winter barley, winter pea, and rapeseed) during seven growing seasons (2012 to 2023) in a well-established (since 2007) alley cropping agroforestry trial site in Southwestern Germany. The trial integrated three different agroforestry practices in a randomized block design: (i) willow short-rotation coppice, (ii) walnut trees for nut production, and (iii) diverse hedgerows. The relationship between crop yield and climatic water balance was analyzed using a linear mixed-model. In this unique long-term comparison, we demonstrate that individual alley cropping practices exhibited distinct yield patterns with increased distance to tree rows. In contrast to the willow short rotation coppice, walnut and hedgerows did not evoke significant winter crop yield declines at proximity. While in the walnut plots yields did not significantly vary with distance to tree rows, yields adjacent to hedge rows declined significantly towards the alley center. Moreover, tree rows contributed to stable crop yields under fluctuating water availability in their proximity and up to the alley center on their leeward side while yields significantly varied with changing climatic water balance on the windward side. Our results underline the potential of agroforestry to sustain yields in the face of increasingly variable water availability, further substantiating the contribution of alley cropping agroforestry for farming systems' resilience to increasingly variable weather conditions. They moreover contribute to planning and policy support for advancing agroforestry as a climate smart solution in temperate regions.

**Keywords** Short rotation coppice · Walnut · Hedge · Climate change adaptation · Crop yield · Climatic water balance · Drought · Tree–crop interaction · Agroecological intensification · Alley cropping

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## 1 Introduction

In the face of climate change, European agriculture is in dire need of adaptation options. Warm season droughts are becoming more frequent in Central Europe (Markonis et al. 2021), already causing severe yield losses (Lüttger and Feike 2018; Brás et al. 2021). This requires the promotion of more resilient and diversified cropping systems, and agroforestry is widely regarded as a key adaptation measure in the face of increasingly extreme weather events (IPCC 2019; van Noordwijk et al. 2021). The regulating effects of woody components on the microclimate have been observed by various studies, namely the reduction of temperature extremes, wind speed, and evaporation (Schöneberger et al. 2012; Kanzler et al. 2019; Swieter et al. 2022; Jacobs et al. 2022). This microclimate amelioration benefits a more climate-resilient agricultural production, among other agroforestry-inherent traits such as biodiversity enhancement, improved soil fertility, and carbon sequestration (Cardinael et al. 2021).

However, competition for light, water, and soil nutrients between arable crops and woody components has been reported to negatively affect crop yields in agroforestry systems (Arenas-Corraliza et al. 2022). In temperate agroforestry systems, light is considered to be the most limiting factor for agricultural production (Ehret et al. 2015). The reduction in photosynthetically active radiation caused by the tree stands has been shown to result in yield and quality decline (sometimes up to 50% yield decrease) in field crops such as various cereals and legumes (Reynolds et al. 2007; Dufour et al. 2013; Artru et al. 2017). Most cultivars of arable crops in temperate climates have been selected for full light conditions. Therefore, it is important to understand how they respond under altered light conditions (Arenas-Corraliza et al. 2022), and to what degree other factors such as seasonal climatic changes and aspect arrangements can influence the net effects of shading (Anderson et al. 2009; Cardinael et al. 2021; Swieter et al. 2022).

While in a tropical, sub-tropical, and Mediterranean context the benefit of agroforestry for agricultural yields has been established for several production systems (Gomes et al. 2020; Arenas-Corraliza et al. 2022) and is projected to be of increasing importance in the face of climate change (Cardinael et al. 2021), there is still a large research gap for temperate climates. Alley cropping — the intentional integration of trees and crops with widely spaced tree rows and in-between crop alleys (Wolz and DeLucia 2018) — is an especially attractive form of agroforestry for temperate climates, as it offers a low-input system (via improved water management and nutrient cycling) for the simultaneous growth of food production and biomass for improved land use (Quinkenstein et al. 2009). Additionally, the windbreak and shade provided

by the tree rows can reduce water loss and improve soil moisture retention due to the lower radiation intensity, reduced wind speed, decreased surface run-off, and more moderate soil temperatures (Bird 1998; Quinkenstein et al. 2009; Jacobs et al. 2022). The effects on air temperature, relative humidity, and evapotranspiration show more varied results, however (Jacobs et al. 2022). Variation among results (yield, water use, microclimate effects, soil effects, etc.) in agroforestry studies is often attributed to differences in tree stand and management (Luedeling et al. 2016; Jacobs et al. 2022). However, our knowledge base on the effects of alley cropping is strongly shaped by research on the management of trees for biomass production, leaving major knowledge gaps when trees are managed for other uses, such as fruit and nut production (Wolz et al. 2018). Many studies in the field identify topography, soil, climate, tree and crop cultivar, canopy management, etc. as the prominent factors influencing microclimate, yield, and water balance (Jacobs et al. 2022; Majaura et al. 2024). Water competition has been shown to vary depending on the tree selection, crops, canopy structure, and of course rainfall patterns (Wang et al. 2021). Furthermore, the spatial distance between the tree rows may play a key role in crop yield and performance due to its influence on competition factors (Quinkenstein et al. 2009; Ivezić et al. 2021). Due to these variable factors, the overall influence that air temperature and relative humidity, together with factors of tree canopy, have on end water use and crop yields is still uncertain and requires more detailed investigation (Grace 1988; Bird 1998; Swieter et al. 2022; Jacobs et al. 2022). Furthermore, despite a growing body of research on the various effects of agroforestry, there remains a significant knowledge gap regarding the variability in yield performance, especially relative to water availability. The heterogeneous effect of tree rows on crop yields in relation to water availability has been acknowledged for decades (Pelton 1967; Kort 1988). However, studies analyzing yield data within alley cropping systems over several years are rare, limiting our understanding of yield dynamics in respect to inter-annual weather variability (Swieter et al. 2022). Although models have been developed to predict tree–crop interactions, yields, ecosystem services, and profitability of agroforestry systems, additional empirical research is needed to improve the accuracy of these models through better parameterization (Van Noordwijk and Lusiana 1999; van der Werf et al. 2007; Graves et al. 2011; Luedeling et al. 2016; Dupraz et al. 2019). Long-term data is necessary to make more robust estimates of the dynamic performances of yields in temperate agroforestry, as here the comparison of dry and wet years is possible, allowing to draw more conclusive relationships between agroforestry practices and yield dynamics across varying weather conditions, namely water surplus and deficit.

This study addresses the uncertainties in the relationship between agroforestry practices and interannual fluctuations in water availability. We do this by examining yield data on a well-established silvoarable agroforestry trial site (established in 2007) at the experimental station Ihinger Hof of the University of Hohenheim in Stuttgart, Germany, one of the oldest, still running agroforestry experiments in Germany (Figure 1). The yield data was analyzed for seven cropping seasons within a 12-year period, from five different winter crops (winter wheat, triticale, winter barley, winter pea, and rapeseed) grown in three alley cropping systems (short-rotation coppice, walnut trees for nut production and diverse autochthonous hedges). This allows for a comprehensive investigation, wherein we assume to see yield pattern differences dependent on the differing tree components within the system and improved yield performance in years with low climatic water balance (CWB). The findings are expected to inform adaptation strategies to climate change through agroforestry practices. Incorporating the CWB into our analysis will elucidate the relationship between agroforestry and water availability, highlighting the potential benefits or drawbacks of these practices in a changing climate.

## 2 Materials and methods

### 2.1 Study site/experimental design

The study site covering 7.1 ha is located at the Agricultural Experimental Station Ihinger Hof of the University of Hohenheim (48.7 °N, 8.9 °E, 475 m asl) in southwestern Germany (Figure 2). The mean annual temperature measured locally 2012–2023 is 9.8 °C and mean annual precipitation amounts to 661 mm. Between May and August, winds are mainly blowing from the west ranging between north- and southwester. The soil is a loess-derived Luvisol with soil textures ranging from loam to clay, with loamy clay

being most prevalent. The alley-cropping agroforestry trial was established in autumn and spring of 2007/2008 and features three different tree row configurations representing the diverse characteristics of temperate alley-cropping practice (Wolz and DeLucia 2018):

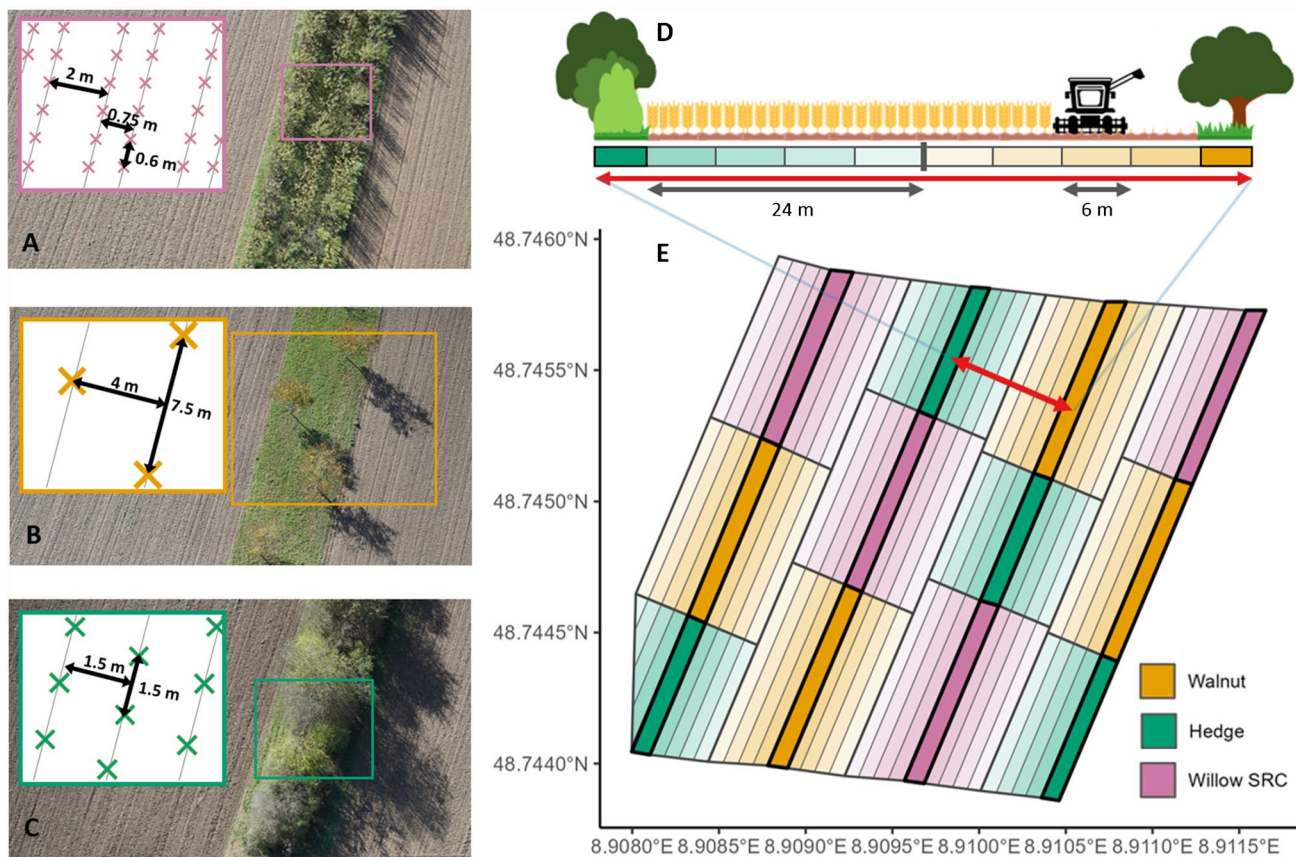
1. Willow short rotation coppice (SRC): willow hybrid clone “Tordis” (*Salix schwerinii* × *viminialis*), planted in three double rows spaced 2 m between each other. Within one double row, single rows are spaced 0.75 m apart, spacing along the row is 0.6 m (Figure 2 A). Distance of tree rows to the crop alley is ~0.9 m. Willows are harvested every 3 years.
2. Walnuts: productive nut-bearing *Juglans regia* variety “Weinheimer (No. 139),” planted in double rows with rows spaced 4 m apart and trees at 7.5 m distance to each other. Distance of tree rows to the crop alley is 2 m (Figure 2B).
3. Autochthonous hedge: features a diverse mix of woody species, typical for their occurrence in the agricultural landscape. The hedge was planted in three parallel rows with a spacing of 1.5 m between the rows and the plants within each row. Distance of hedge rows to the crop alley is 2.5 m (Figure 2C).

The three different alley-cropping agroforestry practices are arranged in a block design with four replicates (Figure 2E). Tree rows are oriented 23.03° clockwise from N (SSW–NNE), 8 m wide and spaced 48 m apart. The area between tree rows, where crops are grown, was divided into eight sections each of 6 m width, thus, reflecting a transect of proximity to tree rows: One plot comprises four sections (0–6, 6–12, 12–18, and 18–24 m distance) east and west of the tree rows of one alley cropping practice as indicated by coloring in Figure 2E. Three plots associated to the same tree row, comprising the three alley cropping practices, form one block. In total, the trial features four blocks and 12 plots.

**Fig. 1** Impression of the agroforestry alley cropping trial with walnut trees, short rotation coppice, and diverse hedges established at the experimental station Ihinger Hof, southwest Germany, in autumn and spring of 2007/2008. The photo was taken at 10 am on 22 September 2022 (Photocredit: Veronika Geisler).







**Fig. 2** The alley cropping experimental site Ihinger Hof. Tree row composition and spacing within the three alley cropping practices are shown in panels A–C. Crosses are example demarcations highlighting tree spacing. Arable alleys are shown in panel D as well as their

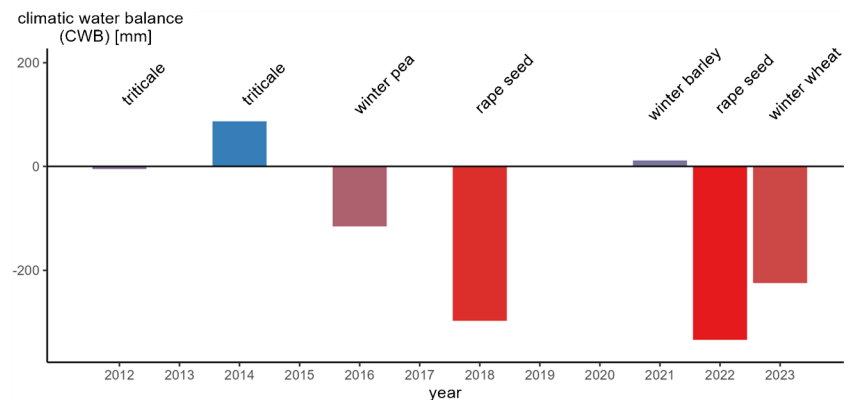
configuration in a block design (E). Plot boundaries are delineated in bold gray, all plots associated to one tree row collectively form a block. Panel D was designed using icons made by Freepik, Georgy, and Aranagraphics from [www.flaticon.com](http://www.flaticon.com).

## 2.2 Data

We used yield data from five different arable winter crops: rapeseed (*Brassica napus* L.), triticale (× *Triticosecale* Wittmack), winter barley (*Hordeum vulgare*), winter pea (*Pisum sativum* L.), and winter wheat (*Triticum aestivum* L.), grown on the entire trial site in 7 years between 2012 and 2023

(see Figure 3). Crop management including application of fertilizer, herbicide, and fungicide was carried out uniformly across the trial site. Yield data was collected over the entire trial site by a GPS-tracked New Holland harvester model CR960 continuously measuring mass flow, providing data with a resolution of 6 m perpendicular to the tree row × 1 m along the alley (Figure 2D). Only data points from between

**Fig. 3** Distribution of climatic water balance (CWB) between May and August between 2012 and 2023 and the studied winter crops at their year of harvest. The displayed winter crop and CWB data were used for analysis.



tree rows and to the west of the first tree row were considered for analysis to avoid confounding yield depression effects of the headland. In total, 28,174 data points were used for the analysis.

To analyze the relationship between yields in agroforestry and annual weather conditions, we used data from the weather station at the Ihinger Hof, located 1 km to the east of the site. Incorporating water–energy interactions, water-balance-based variables are proposed as meaningful estimates of the hydrologic and energetic environment experienced by plants (Stephenson 1990; Fisher et al. 2011). For this purpose, CWB was calculated as the difference between precipitation and potential evapotranspiration from late spring to summer (between May and August) (Figure 3). The time period was selected as relevant for the assessment of tree row effects on crop yields with respect to the timing of tree leaf development in spring and sensitivity of crops to drought in summer (Gobin 2012).

We used the number of growing seasons undergone by the trees as a regressor variable to analyze yield dynamics in relation to the age of the agroforestry system. In the case of the willow SRC, growing seasons were counted with the last harvest of woody biomass as a respective starting point (2012, 2016, 2019, and 2022).

### 2.3 Statistical analysis

We used a mixed modelling approach to analyze spatial yield dynamics across winter crops within the three different alley cropping practices and their interaction with CWB. Multi-level modelling is proposed as a feasible approach to address spatial and temporal complexity which challenges statistical analysis of agroforestry trials (Golicz et al. 2023). Crop yield was used as the response variable and collectively analyzed across all years studied. Blocks (BLOCK) were modelled as a fixed effect and plots (PLOT) as random effects. The different agroforestry practices (TREAT) were integrated as a fixed effect. Moreover, the year (YEAR) effect and its interactions with tree row, alley cropping practice, and distance were modelled as random effects. To model the influence of changing CWB on yields in relation to alley cropping practice, distance to tree rows, and aspect, CWB was added to the model as a fixed effect with linear and quadratic terms as well as their interaction with alley cropping practice, aspect (ASPECT), i.e., crops grown east or west from the tree row, and distance classes (DIST) 0–6, 6–12, 12–18, and 18–24 m from the tree row. Prior to modelling, CWB was divided by 100 to better match its scale to the remaining variables. Additionally, both the crop species (CROP) and age (AGE) were used as fixed covariates in order to distinguish their effect from that of CWB.

The final model is structured as follows:

Response: YIELD ~

Fixed: BLOCK + CROP + TREAT + DIST + ASPECT + TREAT:DIST + TREAT:ASPECT + DIST:ASPECT + CROP:DIST + CROP:ASPECT + TREAT:ASPECT:DIST +

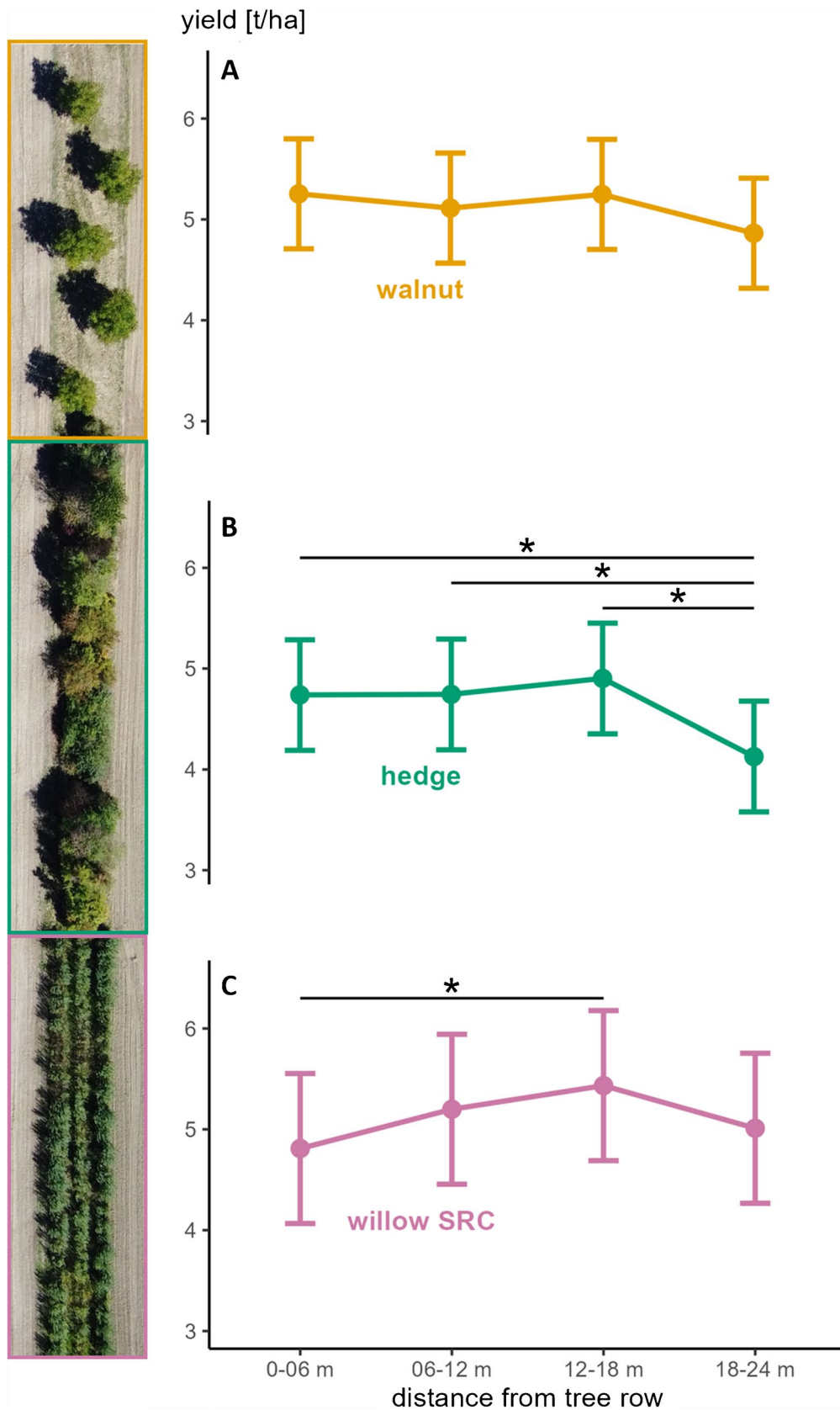
AGE + CWB + I(CWB^2) + TREAT:CWB + TREAT:I(CWB^2) + DIST:CWB + DIST:I(CWB^2) + ASPECT:CWB + ASPECT:I(CWB^2) + TREAT:DIST:CWB + ASPECT:DIST:CWB + TREAT:ASPECT:CWB + TREAT:ASPECT:DIST:CWB  
Random intercept: (1|YEAR) + (1|BLOCK:YEAR) + (1|TREAT:YEAR) + (1|DIST:YEAR) + (1|ASPECT:YEAR) + (1|TREAT:DIST:ASPECT:YEAR) + (1|BLOCK:PLOT) + (1|BLOCK:PLOT:YEAR)

Models were built with the glmmTMB-R-package (Brooks et al. 2024).

The type II Wald chi-squared test was used for analysis of variance. Respecting marginality (Piepho and Edmondson

**Table 1** Effects of agroforestry variants, distance to trees, aspect, and climatic water balance on winter crop yields in alley cropping agroforestry. *Chisq*, chi-squared; *Df*, degrees of freedom; *Pr(>Chisq)*, *p*-value, probability of obtaining greater chi-squared value than presented; \*significant at alpha = 0.05.

| Predictor             | Chisq    | Df | Pr(>Chisq) |
|-----------------------|----------|----|------------|
| BLOCK                 | 4.2406   | 3  | 0.2366     |
| CROP                  | 499.2029 | 4  | <0.0001*   |
| TREAT                 | 3.5474   | 2  | 0.1697     |
| DIST                  | 52.1087  | 3  | <0.0001*   |
| ASPECT                | 2.3432   | 1  | 0.1258     |
| AGE                   | 0.0843   | 1  | 0.7716     |
| CWB                   | 3.9405   | 1  | 0.0471*    |
| I(CWB^2)              | 0.4808   | 1  | 0.4881     |
| TREAT:DIST            | 49.7127  | 6  | <0.0001*   |
| TREAT:ASPECT          | 0.2314   | 2  | 0.8907     |
| DIST:ASPECT           | 8.5091   | 3  | 0.0366*    |
| CROP:DIST             | 43.4126  | 12 | <0.0001*   |
| CROP:ASPECT           | 8.9413   | 4  | 0.0626     |
| TREAT:CWB             | 2.3254   | 2  | 0.3126     |
| TREAT:I(CWB^2)        | 3.1547   | 2  | 0.2065     |
| DIST:CWB              | 31.3350  | 3  | <0.0001*   |
| DIST:I(CWB^2)         | 13.1260  | 3  | 0.0044*    |
| ASPECT:CWB            | 4.3822   | 1  | 0.0363*    |
| ASPECT:I(CWB^2)       | 0.0733   | 1  | 0.7866     |
| TREAT:DIST:ASPECT     | 0.3500   | 6  | 0.9992     |
| TREAT:DIST:CWB        | 7.0149   | 6  | 0.3195     |
| DIST:ASPECT:CWB       | 17.5735  | 3  | 0.0005*    |
| TREAT:ASPECT:CWB      | 0.4433   | 2  | 0.8012     |
| TREAT:DIST:ASPECT:CWB | 5.6310   | 6  | 0.4658     |



**Fig. 4** Winter crop yields with distance to tree rows for three different alley cropping practices, featuring walnut trees (A), autochthonous hedge (B) and willow short rotation coppice (SRC) (C). Yields shown are averaged over the five winter crops studied and over both sides of the tree row. Bars indicate 95% confidence intervals of estimated marginal means. An asterisk denotes significant at  $\alpha = 0.05$ .

2018) in the analysis of variance, the statistics of linear terms were not considered where corresponding quadratic terms were present. Subsequently, the emmeans-R-Package (Lenth 2024) was used to conduct Tukey's tests for pairwise post hoc comparisons. Estimated marginal means were calculated for categorical predictor variables and estimated marginal means of linear trends for reporting estimates of the effect of CWB on yields (Searle et al. 1980; Lenth 2024). Moreover, we quantified the explained variance for LMMs via the "rsquaredGLMM" function from the MuMIn-R-package (Bartoń 2023), yielding a marginal  $R^2$  ( $mR^2$ ) and a conditional  $R^2$  ( $cR^2$ ) based on Nakagawa et al. (2017). While the latter quantifies the variance explained by the fixed effect

and the random effect combined, the marginal  $R^2$  solely accounts for the variance explained by the fixed effect.

### 3 Results and discussion

#### 3.1 Effects of alley cropping agroforestry on winter crop yields

Our model on yield dynamics in three different alley cropping agroforestry systems revealed significant differences in yield dynamics among the studied winter crops. With a marginal  $R^2$  of 0.68, the fixed effects explained substantial variation in the analyzed yield data. Adding the random effects resulted in a conditional  $R^2$  of 0.77. We showed a significant variation in yields between arable crops (CROP) and distance to tree rows (DIST) (Table 1). Averaged across variants, yield levels were highest 12–18 m away from the tree rows significantly higher than at 18–24 m, the center of the alley ( $p = 0.009$ ). No significant variation occurred between blocks and agroforestry practices, with aspect and age of the system. Aspect did not have a significant effect on yield levels, neither as a single predictor nor in interaction with different alley cropping practices (TREAT:ASPECT).

Yield dynamics within alley cropping agroforestry systems are shown to be significantly influenced by the proximity of crops to tree rows (Jacobs et al. 2022). Correspondingly, our analysis revealed distance to tree rows to significantly affect winter crop yields in the alley cropping agroforestry trial. However, when comparing the alley cropping practices (willow SRC, walnut, and autochthonous hedge), different patterns emerged (TREAT:DIST, Table 1).

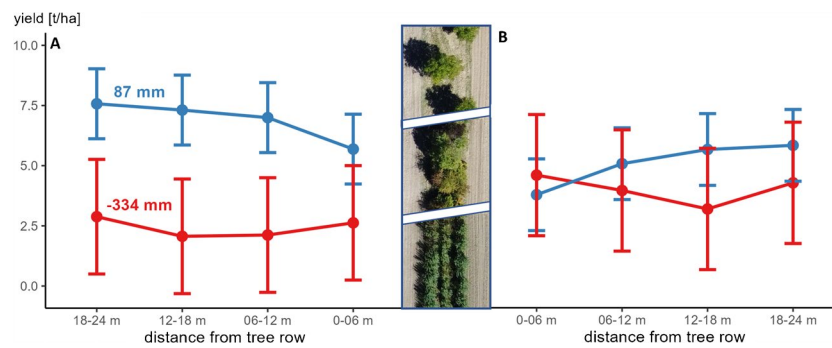
In the willow short rotation coppice plots, yield levels were significantly lower closest to the tree rows (0–6 m) compared to crops at 12–18 m away (Figure 4). This is in line with other studies on yields in alley cropping agroforestry, particularly with SRC (Kanzler et al. 2019; Swieter et al. 2019, 2022). An increase in yield with distance from the tree rows can be attributed to a reducing amount of tree-crop competition for light, nutrients, and water, while the crops still benefit from the sheltering effect of the trees further way (Swieter et al. 2022; Jacobs et al. 2022; Majaura et al. 2024). However, the pattern of increasing yields with distance to tree rows could not be observed for the other two alley cropping practices (Figure 4).

In contrast to the willow SRC system, yield levels directly adjacent to the hedges and walnut trees were not significantly lower compared to the remaining distance classes. While in the walnut system no significant effect of distance to tree rows could be observed, yield levels in the hedge system were lowest at 18–24 m away from the tree rows. The lack of significant yield effect throughout the cropping alley in relation to the walnut tree rows could be attributed

**Table 2** Yield distribution in alley cropping agroforestry for the studied winter crops (triticale (Tr), winter pea (WP), rapeseed (WR), winter barley (WB), and winter wheat (WW)) based on estimated marginal means at different distances from tree rows. Degrees of freedom = 28,085; *emmean*, estimated marginal mean; *SE*, standard error; *lower.CL*, lower confidence level; *upper.CL*, upper confidence level.

| Crop | Distance | emmean | SE    | lower.CL | upper.CL |
|------|----------|--------|-------|----------|----------|
| Tr   | 0–06 m   | 6.859  | 0.505 | 5.870    | 7.848    |
| Tr   | 06–12 m  | 6.199  | 0.505 | 5.209    | 7.190    |
| Tr   | 12–18 m  | 6.330  | 0.505 | 5.339    | 7.320    |
| Tr   | 18–24 m  | 5.578  | 0.506 | 4.587    | 6.569    |
| Tr   | All      | 6.240  | 0.452 | 5.360    | 7.130    |
| WP   | 0–06 m   | 1.797  | 0.213 | 1.379    | 2.215    |
| WP   | 06–12 m  | 1.900  | 0.213 | 1.482    | 2.317    |
| WP   | 12–18 m  | 2.009  | 0.214 | 1.589    | 2.429    |
| WP   | 18–24 m  | 1.935  | 0.214 | 1.515    | 2.356    |
| WP   | All      | 1.910  | 0.192 | 1.530    | 2.290    |
| WR   | 0–06 m   | 3.789  | 1.087 | 1.658    | 5.921    |
| WR   | 06–12 m  | 4.553  | 1.089 | 2.417    | 6.688    |
| WR   | 12–18 m  | 5.294  | 1.089 | 3.160    | 7.429    |
| WR   | 18–24 m  | 3.961  | 1.089 | 1.826    | 6.097    |
| WR   | All      | 4.400  | 0.968 | 2.500    | 6.300    |
| WB   | 0–06 m   | 5.765  | 0.453 | 4.878    | 6.652    |
| WB   | 06–12 m  | 5.063  | 0.453 | 4.174    | 5.952    |
| WB   | 12–18 m  | 4.897  | 0.453 | 4.009    | 5.786    |
| WB   | 18–24 m  | 4.913  | 0.454 | 4.024    | 5.802    |
| WB   | All      | 5.160  | 0.403 | 4.370    | 5.950    |
| WW   | 0–06 m   | 6.458  | 0.513 | 5.452    | 7.464    |
| WW   | 06–12 m  | 7.378  | 0.514 | 6.370    | 8.386    |
| WW   | 12–18 m  | 7.442  | 0.514 | 6.435    | 8.450    |
| WW   | 18–24 m  | 6.948  | 0.514 | 5.941    | 7.956    |
| WW   | All      | 7.060  | 0.457 | 6.160    | 7.950    |





**Fig. 5** Yield distribution in alley cropping agroforestry with changing climatic water balance on both sides of tree rows based on estimated marginal means at maximum (87 mm, blue) and minimum (−334 mm, red) climatic water balance observed in this study. Panel

**A** shows the windward side (west northwest) and panel **B** depicts the leeward side (east southeast). Yields were averaged over the five winter crops studied. Bars indicate 95% confidence intervals of estimated marginal means.

to the relatively wide spacing of trees. In addition, late budding of the species in conjunction with a high crown could further have contributed to a reduced competition for light in comparison to the other systems (Bohn Reckziegel et al. 2022). Like the willow SRC rows, the hedges feature a dense structure, making it surprising that no significant yield declines occurred in the first meters from the tree rows as it was observed for the willow SRC. This might be attributed to an ameliorated sheltering effect provided by the hedge row system which significantly improves the protective microclimatic conditions; as a result, yields on the leeward side of windbreaks are known to benefit most from their sheltering (Pelton 1967; Kort 1988; Baker et al. 2018). Moreover, the heterogeneous structure and tree phenology of the hedge might also contribute to different shading patterns. In addition to sheltering and light competition, other tree crop-interactions could facilitate the contrasting yield distribution between hedge and SRC. In this context, different leaf

litter inputs and competition for nutrients could be relevant (Pardon et al. 2020) since in willow SRC rows nutrients are recurrently exported at harvest. Depending on the conditions in situ, and varieties involved, complex tree–crop interactions result in variable yield responses (Artru et al. 2017; Kanzler et al. 2019; Pardon et al. 2020; Swieter et al. 2022). Existing literature abundantly illustrates that different species, pruning techniques, and management practices significantly affect yield and performance within agroforestry systems, with each system’s unique dynamics leading to varying results (Jacobs et al. 2022). As research on alley cropping agroforestry is strongly focusing on biomass producing systems, however, significant knowledge gaps remain on the effects of alley cropping agroforestry designed for other uses such as fruit and nut production (Wolz et al. 2018). Therefore, the different yield patterns observed in direct comparison between three different alley cropping practices highlight the diverse impact that agroforestry practices can

**Table 3** Model statistics in relation to a change in CWB obtained via the EMMEANS function showing the effect on yields at different distance classes and aspect. CWB, trend shows the estimated yield change in tons per hectare with a pairwise comparison of CWB = 87 mm and CWB = −334 mm. WNW, west northwest; ESE, east southeast; degrees of freedom = 28,085; SE, standard error; lower.CL, lower confidence level; upper.CL, upper confidence level; \*significant at alpha = 0.05.

| Aspect   | Distance | CWB    | SE    | lower.CL | upper.CL | t.ratio | p-value |
|----------|----------|--------|-------|----------|----------|---------|---------|
| WNW      | 0–06 m   | 3.060  | 1.860 | −0.587   | 6.710    | 1.645   | 0.100   |
| WNW      | 06–12 m  | 4.880  | 1.870 | 1.220    | 8.540    | 2.614   | 0.009*  |
| WNW      | 12–18 m  | 5.250  | 1.870 | 1.590    | 8.900    | 2.812   | 0.005*  |
| WNW      | 18–24 m  | 4.690  | 1.870 | 1.040    | 8.350    | 2.515   | 0.012*  |
| WNW      | All      | 4.470  | 1.710 | 1.120    | 7.820    | 2.618   | 0.009*  |
| ESE      | 0–06 m   | −0.815 | 1.950 | −4.650   | 3.020    | −0.417  | 0.677   |
| ESE      | 06–12 m  | 1.110  | 1.960 | −2.720   | 4.950    | 0.569   | 0.570   |
| ESE      | 12–18 m  | 2.470  | 1.960 | −1.360   | 6.310    | 1.263   | 0.207   |
| ESE      | 18–24 m  | 1.560  | 1.960 | −2.280   | 5.390    | 0.796   | 0.426   |
| ESE      | All      | 1.080  | 1.810 | −2.460   | 4.620    | 0.599   | 0.549   |
| Combined | 0–06 m   | 1.120  | 1.610 | −2.040   | 4.290    | 0.697   | 0.486   |
| Combined | 06–12 m  | 3.000  | 1.620 | −0.173   | 6.160    | 1.853   | 0.064   |
| Combined | 12–18 m  | 3.860  | 1.620 | 0.691    | 7.030    | 2.388   | 0.017*  |
| Combined | 18–24 m  | 3.120  | 1.620 | −0.043   | 6.290    | 1.933   | 0.053   |
| Combined | All      | 2.780  | 1.430 | −0.035   | 5.590    | 1.936   | 0.053   |



create on a field level. It is important to properly differentiate between these practices when communicating their effects. This underscores the importance to further investigate the complex tree–crop interactions that shape these effects in different system configurations. Especially the unexpected contrast in yield patterns between willow SRC and hedge rows deserves further investigation in future studies.

Moreover, it needs to be stressed that yield patterns are not stable and vary with changing onsite conditions (Rivest and Vézina 2015; Baker et al. 2018), as explored in the next section. Thus, this mean yield pattern with distance to trees might not reflect yearly response curves.

Estimated mean yield levels also differed significantly between crops. Winter pea showed significantly lower yield levels compared to triticale ( $p = <0.0001$ ), winter barley ( $p = <0.0001$ ), and winter wheat ( $p = <0.0001$ ). Mean yield of winter wheat was also significantly higher than that of rapeseed ( $p = 0.0001$ ). Table 2 shows the estimated marginal mean yield levels of studied winter crops at different distance classes.

### 3.2 Effects of water availability on yield in temperate alley cropping agroforestry

Due to the increased concern of declining yields resulting from more frequent droughts and higher temporal variability of precipitation (Brás et al. 2021), the compounding effect of water availability on yield across years was especially pertinent in the investigated agroforestry systems. Our data shows that CWB influenced yield dynamics in interaction with distance and aspect to trees (DIST:ASPECT:CWB, Table 1), but no significant variation with yield data as a single quadratic predictor. The observed influence of CWB on yields did not significantly vary between tree components: thus, in this section, the alley cropping practices are looked at in composite.

Yield did not significantly vary with CWB adjacent to both sides of the tree rows (0–6 m). Beyond that we see significant variability in yield levels on the windward (WNW) side but not on the leeward (ESE) side. The analysis of estimated marginal means at respective distance classes for CWB further substantiates that (Figure 5, Table 3). Only on the windward side yield substantially declined with low CWB during spring and summer and benefitted from high CWB. Thus, we find that tree rows have a stabilizing effect on yields subjected to inter annual weather variability, both at proximity to tree rows on either side and up to the alley center on the leeward side. This pattern resembles the influence of trees on microclimate through shading close to trees and wind protection further into the field. In an alley cropping system, the tree rows create a gradient of microclimatic effects (Jacobs et al. 2022). In a system of several tree rows, the wind speeds are known to be lowest on the leeward side,

increasing with distance from the tree strip. This protective effect of the tree row creates a shelter that reduces evaporative loss on the soil and improves temperature–soil–water relationships (Aase and Siddoway 1976). Additionally, the lower air temperature in shaded zones allows for reduced transpiration in the crops, which would be especially relevant on hot and sunny days (Brandle et al. 2004; Kanzler et al. 2019). These effects have the potential, then, to materialize as boosted growth performance due to improved water availability over the growing season especially under dry conditions (Rosenberg et al. 1983, Kanzler et al. 2019), which is in alignment with what we see here, where significant drought-induced yield depressions do not occur close to the tree rows and on their leeward side (Figure 5). However, further research combining long-term yield data with microclimatic measurements would help to further elucidate the mechanisms for inter-annual weather effects on yield.

## 4 Conclusion and outlook

In this study, we aimed to elucidate the effects of water availability on winter crop yields in different temperate alley cropping agroforestry practices using high-resolution yield data from a long-term trial in southwestern Germany. As expected, we observe diverging yield patterns between different tree components. However, the sheltering of crops exposed to year to year fluctuation in water availability is consistent across all variants. Agroforestry is widely advocated as an adaptation strategy for enhancing agricultural resilience to extreme weather events caused by climate change. Our findings contribute to understanding long-term yield dynamics within agroforestry practices and under variable water availability. Through this novel direct comparison between alley cropping practices, we were able to show both the variation of yield dynamics between different alley cropping practices and that tree rows buffer the effects of variable weather conditions on winter crop yields. We showed that while not differing in overall yield levels, spatial distribution of yields in alley cropping with walnut trees and diverse hedges features patterns that significantly deviate from the typical transition of low yields close to tree rows to higher yields towards the alley center. In walnut plots, yields did not change significantly with distance to trees and adjacent to hedges, yields dropped towards the alley center. The stability of yields with changing water availability close to tree rows and further on their leeward side is consistent between agroforestry practices. The heterogeneous microclimate of alley cropping agroforestry holds the potential to provide stabilized yields for mitigation of adverse climate change effects. Understanding yield patterns in different alley cropping systems and

their response to changes in CWB can inform targeted management practices such as choice of row distances and cultivar selection to optimize overall performance in the long-term. Additionally, this data can be used to improve the parametrizations of existing prediction models for agroforestry systems in temperate climates. Our long-term data demonstrates that continued research and monitoring are essential for optimizing these systems to maximize yields and develop robust strategies against the backdrop of climate change-induced weather extremes. Future studies integrating on-site microclimatic and edaphic measurements will help to further improve our understanding of tree–crop interactions and differences between different weather conditions and agroforestry practices. This will be addressed by subsequent research efforts at this study site. Moreover, gaining insights on water balances and water use complementarity between trees and annual crops in temperate agroforestry systems would benefit the optimization of climate resilient agroforestry systems and will be a targeted in follow-up investigations. Additionally, measurements of the tree components will be part of upcoming studies analyzing the productivity of the entire agroforestry variants.

**Acknowledgements** We gratefully acknowledge the late Professor Wilhelm Claupen, whose foresight in 2007 to establish the agroforestry field trial at the experimental station Ihinger Hof made this research possible. Special thanks also to all the technicians and supervisors involved from the experimental station of the Ihinger Hof.

**Authors' contributions** A.H.S. and O.K. conceptualized and designed the study. Data preparation, analysis, and visualization were performed by O.K. with contribution of A.H.S. H.P.P. helped with the choice of model for statistical analysis. The first draft of the manuscript was written by J.M. and O.K., and all authors were involved in the review and revision of this manuscript. A.H.S. supervised the study. All authors read and approved the final manuscript.

**Funding** Open Access funding enabled and organized by Projekt DEAL. This work was supported by the Eva Mayr-Stihl Foundation through funding of the project “Coordination Office for Agroforestry Systems Research (kAfo)” and funds of the Federal Ministry of Food and Agriculture (BMEL) 636 based on a decision of the Parliament of the Federal Republic of Germany via the Federal Office for Agriculture and Food (BLE) 637, grant number 2822HUM010 (project “Humax”).

**Data availability** The datasets and code generated during and/or analyzed during the current study are available in the Zenodo repository, <https://doi.org/10.5281/zenodo.15092719>.

## Declarations

**Consent for publication** Not applicable.

**Consent to participate** Not applicable.

**Ethics approval** Not applicable.

**Conflict of interest** The authors declare no competing interests.

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## References

- Aase JK, Siddoway FH (1976) Influence of tall wheatgrass wind barriers on soil drying. *Agron J* 68:627–631. <https://doi.org/10.2134/agronj1976.00021962006800040024x>
- Anderson SH, Udawatta RP, Seobi T, Garrett HE (2009) Soil water content and infiltration in agroforestry buffer strips. *Agrofor Syst* 75:5–16. <https://doi.org/10.1007/s10457-008-9128-3>
- Arenas-Corraliza MG, López-Díaz ML, Rolo V, Cáceres Y, Moreno G (2022) Phenological, morphological and physiological drivers of cereal grain yield in Mediterranean agroforestry systems. *Agric Ecosyst Environ* 340:108158. <https://doi.org/10.1016/j.agee.2022.108158>
- Artru S, Garré S, Dupraz C, Hiel M-P, Blitz-Frayret C, Lassois L (2017) Impact of spatio-temporal shade dynamics on wheat growth and yield, perspectives for temperate agroforestry. *Eur J Agron* 82:60–70. <https://doi.org/10.1016/j.eja.2016.10.004>
- Baker TP, Moroni MT, Mendham DS, Smith R, Hunt MA (2018) Impacts of windbreak shelter on crop and livestock production. *Crop Pasture Sci* 69:785–796
- Bartoń K (2023) MuMIn: multi-model inference (1.47. 5)
- Bird PR (1998) Tree windbreaks and shelter benefits to pasture in temperate grazing systems. *Agrofor Syst* 41:35–54. <https://doi.org/10.1023/A:1006092104201>
- Bohn Reckziegel R, Sheppard JP, Kahle H-P, Larysch E, Spiecker H, Seifert T, Morhart C (2022) Virtual pruning of 3D trees as a tool for managing shading effects in agroforestry systems. *Agrofor Syst* 96:89–104. <https://doi.org/10.1007/s10457-021-00697-5>
- Brandle JR, Hodges L, Zhou XH (2004) Windbreaks in North American agricultural systems. In: Nair PKR, Rao MR, Buck LE (eds) *New vistas in agroforestry: a compendium for 1st World Congress of Agroforestry, 2004*. Springer, Netherlands, Dordrecht, pp 65–78
- Brás TA, Seixas J, Carvalhais N, Jägermeyr J (2021) Severity of drought and heatwave crop losses tripled over the last five decades in Europe. *Environ Res Lett* 16:065012. <https://doi.org/10.1088/1748-9326/abf004>
- Brooks M, Bolker B, Kristensen K, Maechler M, Magnusson A, McGillicuddy M (2024) glmmTMB: generalized linear mixed models using template model builder. *R Packag Vers* 1:9
- Cardinael R, Cadisch G, Gosme M, Oelbermann M, van Noordwijk M (2021) Climate change mitigation and adaptation in agriculture: why agroforestry should be part of the solution. *Agric Ecosyst Environ* 319:107555. <https://doi.org/10.1016/j.agee.2021.107555>
- Dufour L, Metay A, Talbot G, Dupraz C (2013) Assessing light competition for cereal production in temperate agroforestry systems using experimentation and crop modelling. *J Agron Crop Sci* 199:217–227. <https://doi.org/10.1111/jac.12008>
- Dupraz C, Wolz KJ, Lecomte I, Talbot G, Vincent G, Mulia R, Busière F, Ozier-Lafontaine H, Andrianarisoa S, Jackson N, Lawson G, Dones N, Sinoquet H, Lusiana B, Harja D, Domenicano

- S, Reyes F, Gosme M, Van Noordwijk M (2019) Hi-sAFE: a 3D agroforestry model for integrating dynamic tree–crop interactions. *Sustainability* 11. <https://doi.org/10.3390/su11082293>
- Ehret M, Graß R, Wachendorf M (2015) The effect of shade and shade material on white clover/perennial ryegrass mixtures for temperate agroforestry systems. *Agrofor Syst* 89:557–570. <https://doi.org/10.1007/s10457-015-9791-0>
- Fisher JB, Whittaker RJ, Malhi Y (2011) ET come home: potential evapotranspiration in geographical ecology. *Glob Ecol Biogeogr* 20:1–18
- Gobin A (2012) Impact of heat and drought stress on arable crop production in Belgium. *Nat Hazards Earth Syst Sci* 12:1911–1922. <https://doi.org/10.5194/nhess-12-1911-2012>
- Golicz K, Piepho H-P, Minarsch E-ML, Niether W, Große-Stoltenberg A, Oldeland J, Breuer L, Gattinger A, Jacobs S (2023) Highlighting the potential of multilevel statistical models for analysis of individual agroforestry systems. *Agrofor Syst* 97:1481–1489. <https://doi.org/10.1007/s10457-023-00871-x>
- Gomes LC, Bianchi FJJA, Cardoso IM, Fernandes RBA, Filho EIF, Schulte RPO (2020) Agroforestry systems can mitigate the impacts of climate change on coffee production: a spatially explicit assessment in Brazil. *Agric Ecosyst Environ* 294:106858. <https://doi.org/10.1016/j.agee.2020.106858>
- Grace J (1988) 3. Plant response to wind. *Proc Int Symp Wind Technol* 22–23:71–88. [https://doi.org/10.1016/0167-8809\(88\)90008-4](https://doi.org/10.1016/0167-8809(88)90008-4)
- Graves AR, Burgess PJ, Liagre F, Terreaux J-P, Borrel T, Dupraz C, Palma J, Herzog F (2011) Farm-SAFE: the process of developing a plot- and farm-scale model of arable, forestry, and silvoarable economics. *Agrofor Syst* 81:93–108. <https://doi.org/10.1007/s10457-010-9363-2>
- IPCC (2019) Climate Change and Land: an IPCC special report on climate change, desertification, land degradation, sustainable land management, food security, and greenhouse gas fluxes in terrestrial ecosystems
- Ivezić V, Yu Y, van der Werf W (2021) Crop yields in European agroforestry systems: a meta-analysis. *Front Sustain Food Syst* 5:606631
- Jacobs SR, Webber H, Niether W, Grahmann K, Lüttschwager D, Schwartz C, Breuer L, Bellingrath-Kimura SD (2022) Modification of the microclimate and water balance through the integration of trees into temperate cropping systems. *Agric Meteorol* 323:109065. <https://doi.org/10.1016/j.agrformet.2022.109065>
- Kanzler M, Böhm C, Mirck J, Schmitt D, Veste M (2019) Microclimate effects on evaporation and winter wheat (*Triticum aestivum* L.) yield within a temperate agroforestry system. *Agrofor Syst* 93:1821–1841. <https://doi.org/10.1007/s10457-018-0289-4>
- Kort J (1988) 9. Benefits of windbreaks to field and forage crops. *Proc Int Symp Wind Technol* 22–23:165–190. [https://doi.org/10.1016/0167-8809\(88\)90017-5](https://doi.org/10.1016/0167-8809(88)90017-5)
- Lenth R (2024) emmeans: estimated marginal means, aka least-squares means. R Package Version 1(10):3
- Luedeling E, Smethurst PJ, Baudron F, Bayala J, Huth NI, van Noordwijk M, Ong CK, Mulia R, Lusiana B, Muthuri C, Sinclair FL (2016) Field-scale modeling of tree–crop interactions: challenges and development needs. *Agric Syst* 142:51–69. <https://doi.org/10.1016/j.agsy.2015.11.005>
- Lüttger AB, Feike T (2018) Development of heat and drought related extreme weather events and their effect on winter wheat yields in Germany. *Theor Appl Clim* 132:15–29. <https://doi.org/10.1007/s00704-017-2076-y>
- Majaura M, Böhm C, Freese D (2024) The influence of trees on crop yields in temperate zone alley cropping systems: a review. *Sustainability* 16. <https://doi.org/10.3390/su16083301>
- Markonis Y, Kumar R, Hanel M, Rakovec O, Máca P, AghaKouchak A (2021) The rise of compound warm-season droughts in Europe. *Sci Adv* 7:eabb9668. <https://doi.org/10.1126/sciadv.abb9668>
- Nakagawa S, Johnson PCD, Schielzeth H (2017) The coefficient of determination R<sup>2</sup> and intra-class correlation coefficient from generalized linear mixed-effects models revisited and expanded. *J R Soc Interface* 14:20170213. <https://doi.org/10.1098/rsif.2017.0213>
- Pardon P, Mertens J, Reubens B, Reheul D, Coussement T, Elsen A, Nelissen V, Verheyen K (2020) Juglans regia (walnut) in temperate arable agroforestry systems: effects on soil characteristics, arthropod diversity and crop yield. *Renew Agric Food Syst* 35:533–549. <https://doi.org/10.1017/S1742170519000176>
- Pelton WL (1967) The effect of a windbreak on wind travel, evaporation and wheat yield. *Can J Plant Sci* 47:209–214. <https://doi.org/10.4141/cjps67-036>
- Piepho HP, Edmondson RN (2018) A tutorial on the statistical analysis of factorial experiments with qualitative and quantitative treatment factor levels. *J Agron Crop Sci* 204:429–455. <https://doi.org/10.1111/jac.12267>
- Quinkenstein A, Wöllecke J, Böhm C, Grünwald H, Freese D, Schneider BU, Hüttel RF (2009) Ecological benefits of the alley cropping agroforestry system in sensitive regions of Europe. *Sustain Impact Assess Land-Use Policies Sensitive Reg* 12:1112–1121. <https://doi.org/10.1016/j.envsci.2009.08.008>
- Reynolds PE, Simpson JA, Thevathasan NV, Gordon AM (2007) Effects of tree competition on corn and soybean photosynthesis, growth, and yield in a temperate tree-based agroforestry intercropping system in southern Ontario, Canada. *Carbon Sequestration Landsc Ecol West Eur* 29:362–371. <https://doi.org/10.1016/j.ecoleng.2006.09.024>
- Rivest D, Vézina A (2015) Maize yield patterns on the leeward side of tree windbreaks are site-specific and depend on rainfall conditions in eastern Canada. *Agrofor Syst* 89:237–246. <https://doi.org/10.1007/s10457-014-9758-6>
- Rosenberg NJ, Blad BL, Verma SB (1983) Microclimate: the biological environment. John Wiley & Sons
- Schoeneberger M, Bentrup G, de Gooijer H, Soolanayakanahally R, Sauer T, Brandle J, Zhou X, Current D (2012) Branching out: agroforestry as a climate change mitigation and adaptation tool for agriculture. *J Soil Water Conserv* 67:128A. <https://doi.org/10.2489/jswc.67.5.128A>
- Searle SR, Speed FM, Milliken GA (1980) Population marginal means in the linear model: an alternative to least squares means. *Am Stat* 34:216–221. <https://doi.org/10.1080/00031305.1980.10483031>
- Stephenson NL (1990) Climatic control of vegetation distribution: the role of the water balance. *Am Nat* 135:649–670
- Swieter A, Langhof M, Lamerre J, Greef JM (2019) Long-term yields of oilseed rape and winter wheat in a short rotation alley cropping agroforestry system. *Agrofor Syst* 93:1853–1864. <https://doi.org/10.1007/s10457-018-0288-5>
- Swieter A, Langhof M, Lamerre J (2022) Competition, stress and benefits: trees and crops in the transition zone of a temperate short rotation alley cropping agroforestry system. *J Agron Crop Sci* 208:209–224. <https://doi.org/10.1111/jac.12553>
- van der Werf W, Keesman K, Burgess P, Graves A, Pilbeam D, Incoll LD, Metselaar K, Mayus M, Stappers R, van Keulen H, Palma J, Dupraz C (2007) Yield-SAFE: a parameter-sparse, process-based dynamic model for predicting resource capture, growth, and production in agroforestry systems. *Carbon Sequestration Landsc Ecol West Eur* 29:419–433. <https://doi.org/10.1016/j.ecoleng.2006.09.017>
- van Noordwijk M, Coe R, Sinclair FL, Luedeling E, Bayala J, Muthuri CW, Cooper P, Kindt R, Duguma L, Lamanna C, Minang

- PA (2021) Climate change adaptation in and through agroforestry: four decades of research initiated by Peter Huxley. *Mitig Adapt Strateg Glob Change* 26:18. <https://doi.org/10.1007/s11027-021-09954-5>
- Van Noordwijk M, Lusiana B (1999) WaNuLCAS, a model of water, nutrient and light capture in agroforestry systems. In: Auclair D, Dupraz C (eds) *Agroforestry for sustainable land-use fundamental research and modelling with emphasis on temperate and Mediterranean applications: selected papers from a workshop held in Montpellier, France, 23–29 June 1997*. Springer, Netherlands, Dordrecht, pp 217–242
- Wang Z, Wu Y, Cao Q, Shen Y, Zhang B (2021) Modeling the coupling processes of evapotranspiration and soil water balance in agroforestry systems. *Agric Water Manag* 250:106839. <https://doi.org/10.1016/j.agwat.2021.106839>
- Wolz KJ, DeLucia EH (2018) Alley cropping: global patterns of species composition and function. *Agric Ecosyst Environ* 252:61–68. <https://doi.org/10.1016/j.agee.2017.10.005>
- Wolz KJ, Lovell ST, Branham BE, Eddy WC, Keeley K, Revord RS, Wander MM, Yang WH, DeLucia EH (2018) Frontiers in alley cropping: transformative solutions for temperate agriculture. *Glob Change Biol* 24:883–894. <https://doi.org/10.1111/gcb.13986>

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